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Summary of: *Leveraging Machine Learning and Wearables Technology for Sepsis Prediction and Workflow Optimization*

| Introduction & Challenge

Sepsis is a leading cause of in-hospital death (Rhee et al., 2019), and each hour of delayed treatment increases mortality risk by 7.6% (Kumar et al., 2006). A deadly monitoring gap exists because general wards rely on periodic manual checks (Petersen et al., 2016). Continuous wearable sensors paired with supervised machine learning offer a potential solution (Haveman et al., 2022; Moor et al., 2021). Within Denmark's single-payer healthcare system, which currently faces a structural shortage of 4,700 nurses (Birk et al., 2024), this technology might offer outsized marginal return and improve patient care.

Research Question: Can data collected by modern wearable devices be used to predict sepsis onset on hospital wards in Denmark, and which organizational conditions effects implementation viability?

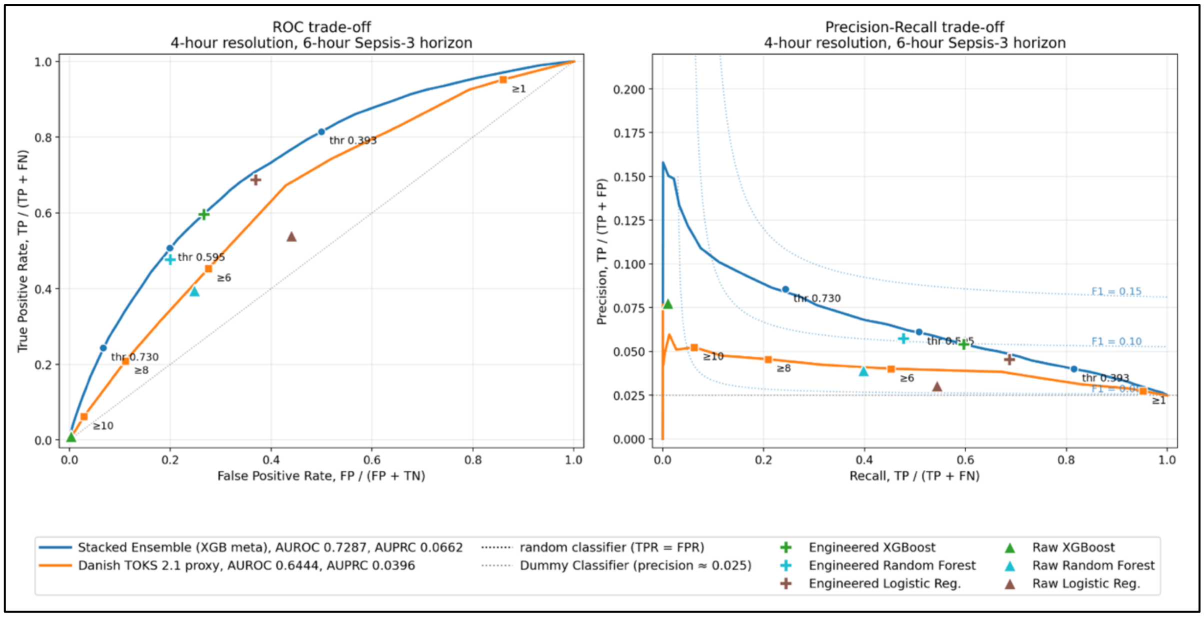
| Dual-Approach

A dual-movement analytical strategy grounded in critical realism (Sayer, 1992) is employes in this paper. The technical movement evaluates predicting Sepsis-3 onset (Singer et al., 2016) using an etiology-specific stacked ensemble constrained to six wearable-proxy vital signs extracted from MIMIC-IV ICU dataset (Johnson et al., 2023). The organizational movement translates these empirical outputs into socio-technical realities, utilizing a heavily contextualized Unified Theory of Acceptance and Use of Technology framework (Venkatesh et al., 2003) to assess the organizational, behavioral, and economic feasibility of ward-level implementation (Holden & Karsh, 2010).

| Technical Findings

As illustrated in Figure 1, the stacked ensemble wearable-proxy estimator scores better at sepsis prediction than the rule-based proxy of current clinical practice, achieving a 67% improvement in sepsis-specific discrimination over the clinical baseline, defined as AUPRC lift over chance. This is when trained on a dataset with vital signs updated ever 4 hours. By increasing monitoring frequency in the training set the stacked ensemble gained another 41% improvement in sepsis-specific discrimination. Suggesting that machine learning and higher monitoring frequency greatly improves the ability to predict sepsis onset.

Figure 1 - Predictive Ability of Estimators



| Organizational Translation

Arguing from pure technological optimism ignores that nurses employ tacit visual cues and holistic concern that algorithms cannot replicate. Viewed through the theory of Hybrid Intelligence (Jarrahi et al., 2022), the algorithm only upgrades the non-human analytical component, while its clinical viability also depends on the nurse's contextual judgment. The nature of the task of predicting the small minority of patients who will suffer from sepsis from limited vital sign signal, inevitably causes a significant False Positive Rate (FPR). Deploying the system to drive clinical decision-making would be detrimental, with binary alarm deployment falling into the automating trap of substituting human judgment with algorithmic execution (Zuboff, 1985), risking severe alarm fatigue (Cvach, 2012). Conversely, an unfiltered informating strategy (Zuboff, 1985) surfacing all 96 complex underlying features risks severe cognitive overload. Because trust in automation is fragile (Parasuraman & Riley, 1997), a system clinicians cannot intuitively grasp triggers trust erosion and behavioral pathologies like disuse or blind obedience. If the cognitive burden negates Effort Expectancy (Venkatesh et al., 2003), the technology becomes organizationally unviable despite its mathematical accuracy.

| Conclusion & Recommendations

The current limit on ward-level detection is the inability of manual tools to transform physiological signals, yet predictive accuracy remains subordinate to the organizational infrastructure that governs its use (Holden & Karsh, 2010). To ensure successful implementation, this paper recommends deploying a 1–3-tiered risk-score interface that mimics existing system, and cognitive routines. Operating primarily in an informating rather than automating mode lowers the risk of alarm fatigue (Zuboff, 1985). To enshure machine->human translation froma Hybrid Intelligence perspective requires empowering clinicians to understand and interrogate the systems reasoning behind it's risk score, building trust and integrating their tacit knowledge (Jarrahi et al., 2022). Finally, phased pilot testing on a single ward is seen as essential to safely calibrate thresholds against real-world sensor noise, establish clinical governance, and build user trust before broader expansion (Parasuraman & Riley, 1997).

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